

Bayesian Filtering for Interaction-Structure Learning in Point-Process Interacting Particle Systems

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Abstract

We develop a Bayesian filtering framework for recovering the interaction structure of a continuous-time Interacting Particle System (IPS) whose components evolve according to coupled point-process dynamics and are observed only through their jump times. Instantiating the framework on a network of stochastic Integrate-and-Fire neurons, we derive the exact posterior law of the interaction matrix W under a partial observation of the spike trains, characterise its dynamics between and at jumps, and establish posterior consistency, with an explicit non-asymptotic CLT-based convergence rate in a tractable two-neuron specialisation (with an exogenous Poisson driver) and a column-wise consistency sketch in the general n -neuron case. We factorise the posterior along incoming edges to reduce the combinatorial burden from $2^{n(n-1)}$ to $n 2^{n-1}$, validate the theory on simulations, and extend the method to continuous-valued interactions via a Sequential Monte Carlo particle filter, in line with recent work on nonparametric inference of interaction laws. The proposed approach complements targeted-learning and EM-based inference methods discussed in the course and provides provable statistical guarantees for learning the structure of interactions in complex systems.

1 Introduction

Interacting Particle Systems (IPS) are the natural mathematical home for the phenomena that underpin this course [Liggett, 2005, Aldous, 2013]. Swarms of animals, agents in a city, synapses in a cortical column, traders in a limit-order book: each of these can be described as a collection of elementary units that evolve stochastically and influence one another through a (typically local) interaction rule. A recurring theme of the lectures is that the resulting collective dynamics—phase transitions, consensus, metastability, rare events—are in large part determined by the *topology and strength of interactions* rather than by idiosyncratic details of the agents. This makes the inverse problem of *inferring the interaction structure from observations of the dynamics* a central one for any data-driven theory of complex systems.

The inverse problem has been attacked along several orthogonal lines. Targeted learning [van der Laan and Rose, 2011] has been applied to recover coupling matrices from equilibrium samples via odds-ratio estimation. Agent-Based Model (ABM) inference treats the parameters as latent variables and uses an Expectation–Maximisation or Moment-Matching strategy on time-aggregated observations [Banisch, 2015]. All of these methods operate either in equilibrium, on coarse-grained observations, or on discrete-time data. In many natural examples—spike trains of neurons, trades in a market, contacts in an epidemic—the observations are instead the *jump times* of a coupled point process, and the identification must be carried out in continuous time.

This paper. We propose a Bayesian filtering framework to address this regime. Our running example is a network of Integrate-and-Fire neurons with binary or real-valued coupling matrix W , a canonical point-process IPS. Under a prior on W we derive closed-form dynamics for the

posterior $p_t^w = \mathbb{P}(W = w \mid \mathcal{F}_t^N)$ given the observed spikes. We prove posterior consistency in the large-time limit and quantify the convergence rate through a Central Limit Theorem on inter-spike log-likelihood ratios. A simple factorisation along incoming edges collapses the support of the posterior from $2^{n(n-1)}$ to $n 2^{n-1}$, which makes the method computationally feasible on moderate network sizes. We validate the theory by simulations and extend the framework to continuous W via a bootstrap Sequential Monte Carlo algorithm, demonstrating on a small example that the filtering perspective survives the discrete-to-continuous transition.

Position in the “Interactions” course. The paper fits within Bloc 3 of the course (inference methods) and, through its continuous-time, spike-train flavour, connects naturally to the point-process lectures of the companion neuroscience programme. Compared to the targeted-learning and EM approaches of Lecture 6, our method requires a stronger assumption—a well-specified generative model—but in exchange yields (i) provable consistency, (ii) explicit non-asymptotic rates, and (iii) access to the full posterior rather than a point estimate, which is useful for uncertainty quantification and active-experiment design. The Bayesian filter may also be viewed as a continuous-time, model-based cousin of causal-discovery procedures [Pearl, 2009, Manzo, 2022]: the posterior on W directly encodes a distribution over directed interaction graphs.

Contributions.

- We formulate the recovery of the interaction matrix W of a point-process IPS as a *nonlinear filtering* problem on the partial observation $\mathcal{F}_t^N$ and derive closed-form evolution equations for the posterior, both between and at jumps.
- We prove posterior consistency in the large-time limit, with an explicit non-asymptotic Berry–Esseen rate via a CLT on the inter-spike likelihood ratios in a tractable two-neuron specialisation and a column-wise consistency sketch for general n .
- We exploit a *factorisation by incoming edges* that reduces the posterior support from $2^{n(n-1)}$ to $n 2^{n-1}$, enabling computation for moderate network sizes.
- We validate the theory by simulations on Erdős–Rényi interaction graphs of sizes $n \in \{2, 3, 5, 7\}$.
- We extend the framework to continuous-valued W via a bootstrap Sequential Monte Carlo algorithm, following the line of Lu et al. [2019].

Outline. Section 2 formulates the model and identifies it as an IPS in the sense of Liggett [2005]. Section 3 derives the filtering equations. Section 4 proves consistency and gives a convergence-rate bound. Section 5 extends the analysis to n neurons with the factorised posterior. Section 6 reports numerical validation. Section 7 describes the SMC extension to continuous interactions. Section 8 connects the approach to targeted-learning, ABM inference and causal inference, and outlines directions for future work.

2 Setup: a point-process IPS with partial observation

2.1 Model

Fix $n \geq 2$ and a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ rich enough to carry all processes below. For each component $i \in \{1, \dots, n\}$ we associate a real-valued hidden state $V_t(i)$ (“potential”) and a counting process $N_t(i)$ (“jumps of i ”). The interaction matrix $W \in \mathcal{M}_n(\mathbb{R})$ is random with prior π_0 and satisfies $W_{ii} = 0$. Conditionally on W , the pair $(V_t, N_t)_{t \geq 0}$ evolves according to

$$dV_t(i) = \sum_{j \neq i} W_{ji} dN_t(j) - V_{t-}(i) dN_t(i), \quad V_0(i) = 0, \quad (1)$$

and $N_t(i)$ has $\mathcal{F}_t$ -stochastic intensity

$$\lambda_t(i) = f(V_{t-}(i)), \quad (2)$$

where $f: \mathbb{R} \rightarrow [0, f_{\max}]$ is a bounded smooth intensity function (e.g. the sigmoid $f(v) = (1 + e^{-v})^{-1}$). Equation (1) reads: at every jump of neighbour j with $W_{ji} \neq 0$, component i integrates the signed contribution W_{ji} ; at every jump of i itself, its state resets to zero. For the binary case used in most of the paper, $W \in \{0, 1\}^{n \times n}$ with $W_{ii} = 0$ and we take $\pi_0 = \text{Ber}(p_{\text{edge}})^{\otimes n(n-1)}$ independent across edges. The continuous extension considered in Section 7 replaces the Bernoulli prior by a Gaussian prior on each W_{ji} .

2.2 An IPS in Liggett's sense

The process $(V_t, N_t)_{t \geq 0}$ is a pure-jump Markov process on the Polish state space $\mathbb{R}^n \times \mathbb{N}^n$ with generator, for any smooth bounded test function φ and state $x = (v, k)$,

$$(\mathcal{A}\varphi)(x) = \sum_{i=1}^n f(v_i) [\varphi(T_i(v), k + e_i) - \varphi(x)], \quad T_i(v)_j = \begin{cases} 0 & \text{if } j = i, \\ v_j + W_{ij} & \text{if } j \neq i, \end{cases} \quad (3)$$

where e_i is the i -th canonical basis vector of $\mathbb{R}^n$. Locality is encoded in the support of W : only the i -th row affects the V -coordinates when i fires, and only the i -th column of W determines the potential of i . With $f \equiv \mathbf{1}_{[0, \infty)}$ and $W_{ji} \in \{0, 1\}$ one recovers a voter-like process on a directed graph [Liggett, 2005]; with f additionally localised on the neighbourhood of a site one obtains a contact-process analogue. Our filtering framework thus applies to a larger family of point-process IPS than integrate-and-fire networks.

2.3 Prior and observation

We observe only the full spike-train filtration $\mathcal{G}_t = \sigma(N_s(i) : s \leq t, i = 1, \dots, n)$ and seek the posterior

$$p_t^w := \mathbb{P}(W = w \mid \mathcal{G}_t), \quad w \in \text{supp}(\pi_0). \quad (4)$$

Neither the potentials V_t nor the prefactor W are observed. Equation (4) is the classical nonlinear filtering problem for a marked point process with unknown parameter [Brémaud, 1981, Snyder, 1972, Kallianpur, 1980], specialised to a discrete (resp. absolutely continuous) prior on W .

3 Bayesian filtering of the interaction matrix

We first state the filter in the simplest non-trivial case $n = 2$ with a binary coupling $J \in \{0, 1\}$, where the posterior reduces to a one-dimensional stochastic process. The general n -neuron treatment is in Section 5.

3.1 Two-neuron model

For tractability we depart slightly from the coupled model of Section 2: instead of letting neuron 2 fire with intensity $f(V_t(2))$, we replace it by an exogenous Poisson driver of fixed rate θ . This isolates the unknown coupling $J = W_{2 \rightarrow 1}$ and yields a one-dimensional posterior; the coupled $n = 2$ case is recovered as the column-wise specialisation of Section 5. Concretely, let $\tilde{N}$ be a Poisson process of rate $\theta > 0$ (exogenous driver) and set

$$dV_t = J d\tilde{N}_t - V_{t-} dN_t, \quad V_0 = 0, \quad \lambda_t(N) = f(V_{t-}). \quad (5)$$

The prior is $\mathbb{P}(J = 1) = 1 - p_0 = 1 - \mathbb{P}(J = 0)$. Write $p_t^1 := \mathbb{P}(J = 1 \mid \mathcal{G}_t)$ and $V_t^{(j)}$ for the potential trajectory one would obtain under hypothesis $J = j$. A Girsanov change-of-measure argument [Brémaud, 1981] yields the filtering equations (sketch in Appendix A):

Proposition 3.1 (Two-neuron filter). *On any interval (τ_k, τ_{k+1}) between consecutive jumps of either $\tilde{N}$ or N , the posterior evolves according to*

$$\frac{dp_t^1}{dt} = p_t^1(1 - p_t^1)[f(V_t^{(0)}) - f(V_t^{(1)})]. \quad (6)$$

At a jump of the exogenous process $\tilde{N}$ the posterior does not change (both hypotheses predict the jump with the same probability θdt); at a jump of N at time τ the posterior is updated by Bayes:

$$p_\tau^1 = \frac{f(V_{\tau-}^{(1)}) p_{\tau-}^1}{f(V_{\tau-}^{(0)})(1 - p_{\tau-}^1) + f(V_{\tau-}^{(1)}) p_{\tau-}^1}. \quad (7)$$

Intuitively, between jumps the posterior drifts towards the hypothesis whose predicted intensity $f(V_t^{(j)})$ is smallest, because the absence of a spike is more consistent with that hypothesis; at a jump the posterior is multiplied by the hypothesis' intensity. Note that this result is obtained as a particular case of the derivation detailed in Section 3.2.

Remark 3.2 (Why only jumps of N matter). In (7) only the observed process N updates the posterior. This is a general feature of point-process filtering: a jump of a process whose intensity depends on the unknown parameter is informative; a jump of a process with parameter-free rate (here $\tilde{N}$ with rate θ) carries information only through the subsequent evolution of the potentials $V_t^{(j)}$.

3.2 Full n -neuron filter

In the n -neuron model of Section 2 each column $W_{\cdot,i}$ of W determines the potential of neuron i but does not appear in any other potential. Under a product prior $\pi_0(W) = \prod_{i=1}^n \pi_0^i(W_{\cdot,i})$, the posterior factorises along columns,

$$p_t^W = \prod_{i=1}^n p_t^i(W_{\cdot,i}), \quad (8)$$

and each factor satisfies an analogue of (6) and (7) with V replaced by the hypothesis-indexed potential $V^{(w)}(i)$. We summarise the resulting dynamics in Proposition 3.3 below and defer the derivation to Section 5.

Proposition 3.3 (n -neuron filter, factorised form). *Let $p_t^i(w)$ be the posterior mass of the configuration $w \in \text{supp}(\pi_0^i)$ on the incoming edges of neuron i . Between jumps,*

$$\frac{dp_t^i(w)}{dt} = p_t^i(w) [\bar{\lambda}_t(i) - f(V_t^{(w)}(i))], \quad \bar{\lambda}_t(i) := \sum_{w'} p_t^i(w') f(V_t^{(w')}(i)), \quad (9)$$

and at a jump of neuron i at time τ , only the factor of neuron i is updated:

$$p_\tau^i(w) \propto f(V_{\tau-}^{(w)}(i)) p_{\tau-}^i(w). \quad (10)$$

Jumps of neighbours $j \neq i$ do not modify p_t^i directly but do update the hypothesis-indexed potentials $V_t^{(w)}(i)$.

Remark 3.4 (Consistency with the two-neuron case). For $n = 2$ the support $\text{supp}(\pi_0^1) = \{0, 1\}$ and (9) reduces, after writing $p := p_t^1(1) = p_t^1$ and $1 - p = p_t^1(0)$, to $dp/dt = p[\bar{\lambda} - f(V^{(1)})]$ with $\bar{\lambda} = (1 - p)f(V^{(0)}) + pf(V^{(1)})$, which simplifies to $p(1 - p)[f(V^{(0)}) - f(V^{(1)})]$, exactly (6).

The key complexity reduction is that $\text{supp}(\pi_0^i) \subset \{0, 1\}^{n-1}$ has cardinality 2^{n-1} rather than $2^{n(n-1)}$, so the full posterior is stored and updated in $O(n \cdot 2^{n-1})$ memory. This is the observation that unlocks the experiments of Section 6 for n up to 7.

4 Posterior consistency and convergence rate

We now show that the posterior of Proposition 3.1 concentrates on the true value $J^\star$ of J as $t \rightarrow \infty$, and we quantify the rate of convergence in the number of observed spikes.

4.1 Identifiability via Kullback–Leibler

Denote by $r_t := p_t^1 / (1 - p_t^1)$ the posterior odds ratio and let τ_k be the time of the k -th spike of N (we index by spikes of N rather than by generic events because only these reset the potentials). A direct computation from (6)–(7) gives

$$\log r_{\tau_k} = \log r_0 + \sum_{\ell=1}^k X_\ell, \quad X_\ell := \log \frac{K_1(\xi_\ell)}{K_0(\xi_\ell)}, \quad (11)$$

where $\xi_\ell := (V_{\tau_{\ell-1}:\tau_\ell}^{(0)}, V_{\tau_{\ell-1}:\tau_\ell}^{(1)}, \tilde{N}_{[(\tau_{\ell-1}, \tau_\ell])})$ is the full inter-spike trajectory on $(\tau_{\ell-1}, \tau_\ell]$ —including every jump of the exogenous driver $\tilde{N}$ that occurs in that window—and K_j is its law under hypothesis $J = j$. Since $V_{\tau_\ell}^{(j)} = 0$ at every spike of N and $\tilde{N}$ is a homogeneous Poisson process independent of the past, the strong Markov property ensures that the ξ_ℓ form an i.i.d. sequence under each K_j . Consequently (11) is a random walk with drift

$$\mathbb{E}_{J^\star}[X_\ell] = \begin{cases} \text{KL}(K_1 \| K_0) > 0 & \text{if } J^\star = 1, \\ -\text{KL}(K_0 \| K_1) < 0 & \text{if } J^\star = 0, \end{cases} \quad (12)$$

the two KL divergences being strictly positive whenever $f(0) \neq f(1)$ and $\theta > 0$: the first condition ensures that the two hypotheses give different intensities after at least one jump of $\tilde{N}$, and $\theta > 0$ ensures that such a jump occurs with positive probability before the next spike of N . Together these are our identifiability conditions. The Strong Law of Large Numbers applied to (11) therefore yields *posterior consistency*:

$$p_t^{J^\star} \xrightarrow[t \rightarrow \infty]{\text{a.s.}} 1. \quad (13)$$

4.2 CLT-based convergence rate

A more refined analysis yields the speed at which the posterior concentrates. Define, for $\varepsilon \in (0, 1/2)$, the hitting time in the event counting

$$k_\varepsilon := \inf\{k \in \mathbb{N} : |p_{\tau_k}^1 - J^\star| < \varepsilon\}. \quad (14)$$

Let $\mu := \mathbb{E}[X_1 \mid J^\star]$ and $\sigma^2 := \text{Var}[X_1 \mid J^\star]$. By the central limit theorem applied to (11), the rescaled $\log r_{\tau_k} - k\mu$ is asymptotically Gaussian with variance $k\sigma^2$. A Berry–Esseen-type approximation gives the following bound (see Appendix B):

Proposition 4.1 (Non-asymptotic convergence rate). *Assume $J^\star = 1$, take a uniform prior $p_0 = 1/2$ (so that $\log r_0 = 0$), and assume the intensity is bounded both above and below: $0 < f_{\min} \leq f \leq f_{\max}$ (which guarantees $\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$ since $\log f$ stays bounded). Write $A_\varepsilon = \log((1 - \varepsilon)/\varepsilon)$. Then for every $k \geq 1$,*

$$\mathbb{P}(k_\varepsilon \leq k) \geq \Phi\left(\frac{k\mu - A_\varepsilon}{\sigma\sqrt{k}}\right) - \frac{C\rho}{\sigma^3\sqrt{k}}, \quad (15)$$

where Φ is the standard normal cdf and C is the absolute constant from the Berry–Esseen inequality. Equivalently, as $\varepsilon \downarrow 0$, the smallest k guaranteeing $\mathbb{P}(k_\varepsilon \leq k) \geq 1 - \delta$ satisfies

$$k = \frac{A_\varepsilon}{\mu} + z_{1-\delta} \frac{\sigma\sqrt{A_\varepsilon/\mu}}{\mu} + o(\sqrt{A_\varepsilon}),$$

so the leading term A_ε/μ is a Kullback–Leibler commitment time and the $O(\sqrt{A_\varepsilon})$ correction is a Gaussian fluctuation.

We validate Proposition 4.1 numerically in Section 6.

5 Extension to n neurons and factorised posterior

We now give the n -neuron generalisation of Proposition 3.1 together with the factorisation that underlies the algorithm used in the experiments. We start by deriving the general equation driving the posterior between jumps.

5.1 General posterior ODE

In the general case, we consider a network of neurons whose potentials are driven by the following SDE:

$$dV_t^i = -V_{t-}^i dN_t^i + \sum_{j \neq i} w_{ji} dN_t^j$$

where N_t^i is a point process with predictable intensity $f(V_{t-}^i(w))$ associated to neuron i and w is the connectivity matrix. The Girsanov likelihood in this general case is given by

$$L_t(w) = \prod_i \prod_{k: \tau_k^i \leq t} f(V_{\tau_k^i-}^i(w)) \exp\left(-\int_0^t f(V_s^i(w)) ds\right)$$

Finally, the change of measure formula for Poisson processes yields the explicit formula for the posterior: $p_t(w) = \frac{L_t(w)p_0(w)}{\mathbb{E}_{p_0}[L_t(w)]} = \frac{q_t(w)}{\sum_{w'} q_t(w')}$ with prior p_0 over connectivity matrices.

Differentiating this formula in time between spikes yields the ODE satisfied by the posterior:

$$\begin{aligned} \dot{p}_t(w) &= \frac{\dot{q}_t(w) \sum_{w'} q_t(w') - q_t(w) \sum_{w'} \dot{q}_t(w')}{(\sum_{w'} q_t(w'))^2} \\ &= \frac{-q_t(w) \sum_i f(V_t^i(w)) \sum_{w'} q_t(w') + \sum_{w'} q_t(w') \sum_i f(V_t^i(w')) q_t(w)}{(\sum_{w'} q_t(w'))^2} \\ &= \frac{-q_t(w) \Lambda_t(w) Z_t + \sum_{w'} q_t(w') \Lambda_t(w') q_t(w)}{Z_t^2} \\ &= \frac{q_t(w)}{Z_t^2} \left(-\Lambda_t(w) Z_t + \sum_{w'} q_t(w') \Lambda_t(w') \right) \\ &= \frac{q_t(w)}{Z_t} \left(-\Lambda_t(w) + \sum_{w'} \frac{q_t(w')}{Z_t} \Lambda_t(w') \right) \\ &= p_t(w) \left(\sum_{w'} p_t(w') \Lambda_t(w') - \Lambda_t(w) \right) \\ &= p_t(w) (\mathbb{E}_{p_t}[\Lambda_t(w)] - \Lambda_t(w)) \end{aligned}$$

Specifying this formula in the two-neuron case recovers the earlier formula of Section 3.1. Note that this formula extends naturally to real-valued w .

At spikes of neuron i , the posterior update stems directly from the change of measure formula:

$$p_{\tau_k^i}^i(w) = \frac{f(V_{\tau_k^i-}^i(w)) p_{\tau_k^i-}^i(w)}{\mathbb{E}_{p_{\tau_k^i-}^i}[f(V_{\tau_k^i-}^i(w'))]}$$

5.2 Full and factorised posterior

Let π_0 be a prior on $W \in \{0,1\}^{n \times n}$ with $W_{ii} = 0$ almost surely. The naive posterior $p_t^w = \mathbb{P}(W = w \mid \mathcal{G}_t)$ is supported on $2^{n(n-1)}$ configurations. The following observation, central to our method, makes the problem tractable.

Lemma 5.1 (Column-wise factorisation). *Under (1), the potential $V_t(i)$ depends on W only through its i -th column $W_{\cdot,i}$, so the likelihood of the observation $\mathcal{G}_t$ factorises as*

$$\mathcal{L}(\mathcal{G}_t \mid W) = \prod_{i=1}^n \mathcal{L}_i(N(i) \mid W_{\cdot,i}), \quad (16)$$

where $\mathcal{L}_i$ is the likelihood of the spike train of neuron i conditional on the incoming edges $W_{\cdot,i}$. Consequently, if $\pi_0 = \prod_i \pi_0^i$ is a product prior, the posterior is also a product and (8) holds.

Proof. Integrating (1) over any interval between jumps of $N(i)$ gives $V_t(i) = \sum_{j \neq i} W_{ji} \Delta N_t^{(i)}(j)$ where $\Delta N^{(i)}(j)$ counts the spikes of j since the last reset of i . Hence $V(i)$ (and thus $\lambda(i) = f(V(i))$) only involves $W_{\cdot,i}$. The likelihood of a point process with compensator $\int_0^t \lambda_s ds$ factors as $\exp(-\int \lambda) \prod f(V_{\tau-})$ (see, e.g., Thm 2 of Chapter II.3 of Brémaud, 1981), giving (16). Bayes' rule then yields (8). $\square$

Lemma 5.1 cuts the memory from $2^{n(n-1)}$ to $n \cdot 2^{n-1}$. The derivation of Proposition 3.3 is obtained by applying the two-neuron filter within each column separately, the only subtle point being the propagation of the hypothesis-indexed potentials $V_t^{(w)}(i)$: at a jump of neighbour j at time τ , every candidate w has its potential incremented by w_j (the bit of w at position j). This is exactly the book-keeping carried out in the simulator of Section 6.

5.3 Consistency (sketch)

The consistency argument of Section 4.1 extends columnwise. By Lemma 5.1, the posterior factorises along columns, so it is enough to prove consistency separately for each column. Fix neuron i and let $w^* \in \{0,1\}^{n-1}$ be the true incoming-edge configuration of neuron i (i.e. the i -th column of W^*). Write $q_t^i(w) := L_t^i(w) p_0^i(w)$ for the unnormalised posterior mass of column w on neuron i , where L_t^i is the per-column likelihood of Lemma 5.1, and let $r_t^{w,w^*}(i) := \log \frac{q_t^i(w)}{q_t^i(w^*)}$ denote the log-likelihood ratio between w and w^* along the spike train of neuron i . A direct computation gives

$$r_t^{w,w^*}(i) = \int_0^t \log \frac{f(V_s^i(w))}{f(V_s^i(w^*))} dN_s^i + \int_0^t [f(V_s^i(w^*)) - f(V_s^i(w))] ds + \log \frac{p_0^i(w)}{p_0^i(w^*)}, \quad (17)$$

where p_0^i is the marginal prior on the i -th column.

By the Doob–Meyer decomposition of N^i under $\mathbb{P}_{w^*}$ (whose compensator is $\int_0^t f(V_s^i(w^*)) ds$), we can rewrite

$$r_t^{w,w^*}(i) = \int_0^t \left[\log \frac{f(V_s^i(w))}{f(V_s^i(w^*))} f(V_s^i(w^*)) + f(V_s^i(w^*)) - f(V_s^i(w)) \right] ds + \log \frac{p_0^i(w)}{p_0^i(w^*)} + M_t,$$

where M_t is a martingale whose predictable quadratic variation $\langle M \rangle_t$ is bounded above by Ct (since f is bounded above and $\log f$ is bounded on the relevant range). By the strong law of large numbers for square-integrable martingales (Kolmogorov-type, since $\langle M \rangle_t/t^2 \rightarrow 0$), $M_t/t \rightarrow 0$ a.s. Dividing by t , the contributions M_t/t and $\frac{1}{t} \log p_0^i(w)/p_0^i(w^*)$ vanish as $t \rightarrow \infty$, leaving

$$H_t = \frac{1}{t} \int_0^t \left[\log \frac{f(V_s^i(w))}{f(V_s^i(w^*))} f(V_s^i(w^*)) + f(V_s^i(w^*)) - f(V_s^i(w)) \right] ds.$$

The integrand has the form $\Phi(x, y) = y \log \frac{x}{y} + y - x$ with $x = f(V_s^i(w))$ and $y = f(V_s^i(w^*))$. One checks that $\Phi(x, y) \leq 0$ for all $x, y > 0$, with equality iff $x = y$. Thus H_t accumulates strictly negative mass on every time interval where $f(V_s^i(w)) \neq f(V_s^i(w^*))$.

To turn this into a divergence of $r_t^{w, w^*}(i)$ to $-\infty$ for $w \neq w^*$, it suffices to exhibit events on which the two potentials disagree and to show that such events occur a positive fraction of the time. Pick any j with $w_j \neq w_j^*$ and define, for $\delta > 0$ and $k \geq 1$, the event E_k^δ as the conjunction of:

- (A) neuron i does not fire in $[\tau_k^i, \tau_k^i + \delta]$;
- (B) exactly one neighbour from $\mathcal{D} := \{j' : w_{j'} \neq w_{j'}^*\}$ fires in $[\tau_k^i, \tau_k^i + \delta/2]$ (call it j);
- (C) no neighbour from $\mathcal{D}$ fires in $(\tau_k^i + \delta/2, \tau_k^i + \delta]$.

At time τ_k^i , neuron i fires, so $V_{\tau_k^i+}^i(w) = V_{\tau_k^i+}^i(w^*) = 0$ for *every* configuration (the reset is parameter-independent). Let σ denote the time of the unique spike from $\mathcal{D}$ in $[\tau_k^i, \tau_k^i + \delta/2]$ (whose existence is guaranteed by (B), and whose uniqueness rules out cancellations between disagreeing neighbours). Just after σ , the two potentials differ by exactly $w_j - w_j^* \neq 0$. By (C), no further spike from $\mathcal{D}$ occurs in $(\sigma, \tau_k^i + \delta]$; spikes of indices outside $\mathcal{D}$ add the same value to both potentials and hence preserve the gap. Combined with (A) (no reset of neuron i), this implies $V_s^i(w) - V_s^i(w^*) = w_j - w_j^* \neq 0$ for all $s \in [\sigma, \tau_k^i + \delta]$, hence $f(V_s^i(w)) \neq f(V_s^i(w^*))$ for a duration at least $\delta/2$. The same construction extends, with minor modifications, to the real-valued case.

Under the standing assumption $f \leq f_{\max}$, the probability that neuron i does not fire in a window of length δ is at least $\exp(-f_{\max}\delta) > 0$, and the analogous Poisson-rate bounds for neuron j give $\mathbb{P}(E_k^\delta) \geq c(\delta) > 0$. Provided f is in addition bounded below by some $f_{\min} > 0$ (so that the inter-spike times of N^i have finite mean), the strong Markov property at the spikes of N^i ensures that the average number of such events up to time t grows linearly in t by the law of large numbers. Combining this with the strict negativity of the integrand on E_k^δ yields $H_t \rightarrow -c < 0$ a.s., hence $r_t^{w, w^*}(i) \rightarrow -\infty$ a.s. for every $w \neq w^*$, and therefore $p_t^i(w^*) \rightarrow 1$ a.s. A union bound over the finitely many $w \in \{0, 1\}^{n-1} \setminus \{w^*\}$ preserves almost sure convergence; combined with the column-wise factorisation (8), this gives consistency of the full posterior.

Remark 5.2 (Sigmoid intensity). The sigmoid $f(v) = (1 + e^{-v})^{-1}$ used in our experiments tends to 0 as $v \rightarrow -\infty$ and therefore violates the $f \geq f_{\min} > 0$ assumption underlying the LLN step above. In practice the resets at spikes of neuron i keep V^i in a bounded regime, and the empirical results of Section 6 confirm that consistency holds. A fully rigorous treatment for unbounded-below intensities would require an ergodicity assumption on V^i in the spirit of Cormier et al. [2020] and is the subject of ongoing work.

6 Numerical experiments

We now test the filtering framework on simulated data. All experiments use the sigmoid intensity $f(v) = (1 + e^{-v})^{-1}$, thinning upper bound $f_{\max} = 1$, and the exact algorithm implied by Proposition 3.3: between jumps we integrate the ODE (9) with an Euler scheme of step $dt = 10^{-2}$ (checked to be stable); at each accepted spike of neuron i we apply the multiplicative update (10). Source code (in Python, NumPy only) and commands to reproduce every figure are provided in the companion repository.

6.1 Two-neuron posterior trajectories

Figure 1 shows 20 Monte-Carlo trajectories of the posterior $p_t^1 = \mathbb{P}(J = 1 \mid \mathcal{G}_t)$ in the two-neuron model of Section 3.1, for $J^* \in \{0, 1\}$, with uniform prior $p_0 = 1/2$ and $\theta = 1$. The pointwise

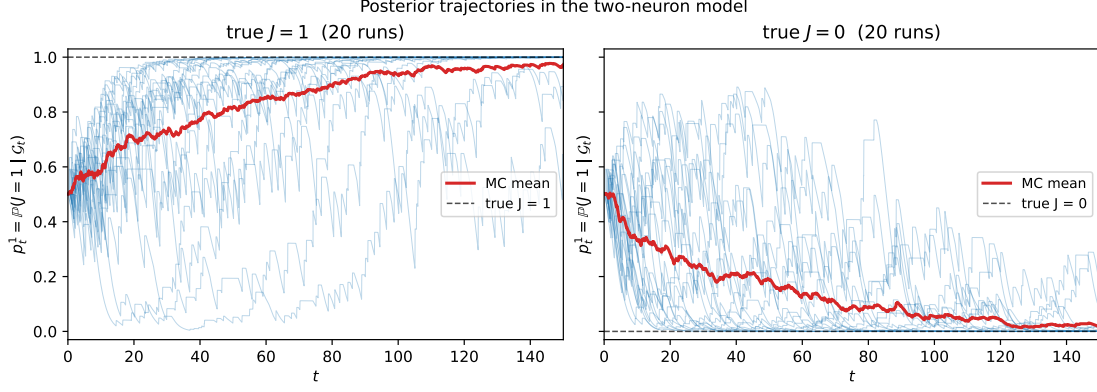


Figure 1: Posterior trajectories $p_t^1 = \mathbb{P}(J = 1 \mid \mathcal{G}_t)$ in the two-neuron model. Left: ground truth $J^* = 1$; right: $J^* = 0$. Grey-blue lines are $N = 20$ individual MC replicates; red is their pointwise mean. Convergence to the truth is almost sure by (13).

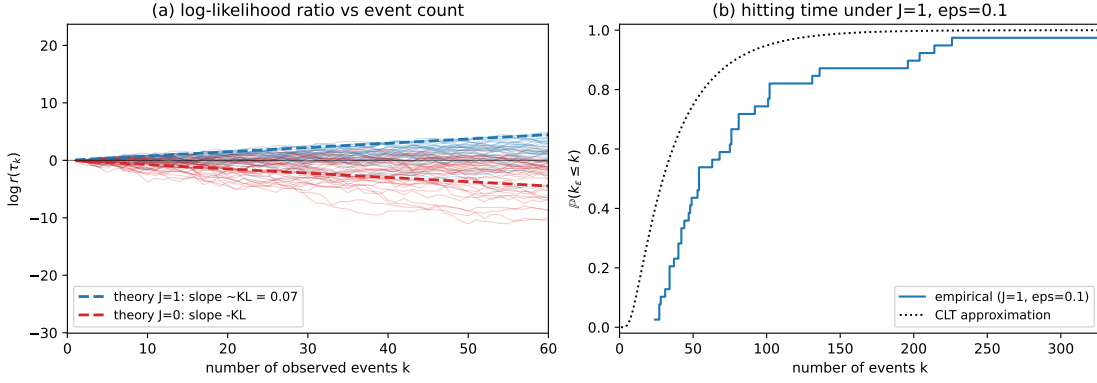


Figure 2: (a) Log-likelihood ratio $\log r(\tau_k)$ versus event count for $N = 40$ replicates under $J^* = 1$ (blue) and $J^* = 0$ (red), together with the theoretical slopes $\pm \text{KL}(K_1 \| K_0)$ (dashed). (b) Empirical CDF of the hitting time k_ε (step, blue) with $\varepsilon = 0.1$ versus the CLT approximation of Proposition 4.1 (dotted, black).

average over trajectories (red) converges to the ground-truth J^* within about $t \approx 100$ time units, in agreement with (13). The individual runs display the expected path-dependence: trajectories that spike early (hence aggregate likelihood mass quickly) commit early to the truth, while rare runs that spike late exhibit a long plateau near the prior before eventually converging.

6.2 Empirical convergence rate

Figure 2 validates Proposition 4.1. Panel (a) plots the log posterior-odds $\log r(\tau_k)$ as a function of event count k for $N = 40$ replicates under each hypothesis. The dashed lines show the predicted drifts $\pm \text{KL}(K_1 \| K_0)$, with KL estimated on 5000 i.i.d. inter-event trajectories. The empirical slopes match the theoretical ones to within Monte-Carlo noise. Panel (b) compares the empirical cumulative distribution of the hitting time k_ε (with $\varepsilon = 0.1$) to the CLT approximation of Proposition 4.1; agreement is excellent despite the modest number of replicates. Numerical estimates gave $\hat{\mu} \approx 0.074$ and $\hat{\sigma}^2 \approx 0.103$, giving a characteristic commitment scale of $k^* \approx A_\varepsilon / \hat{\mu} \approx 30$ spikes, in line with the figure.

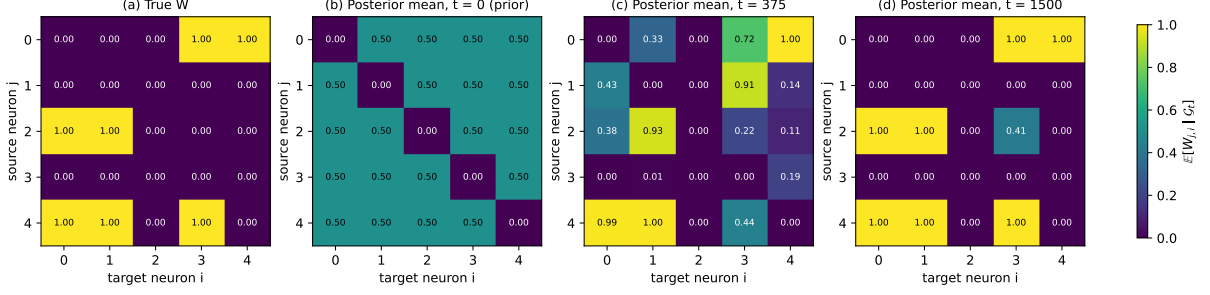


Figure 3: Recovery of W for $n = 5$. (a) Ground truth. (b) Uniform prior at $t = 0$. (c) Posterior mean at $t = T/4$. (d) Posterior mean at $t = T$. Numbers inside each cell are $\mathbb{E}[W_{ji} | \mathcal{G}_t]$.

6.3 Multi-neuron recovery

We next run the factorised filter on a network of $n = 5$ neurons with W^* sampled from $\text{Ber}(0.4)$ i.i.d. per edge. Figure 3 shows the ground-truth coupling matrix, the prior (uniform $1/2$), and the posterior mean $\mathbb{E}[W_{ji} | \mathcal{G}_t]$ at intermediate time $t = T/4$ and at the final time $T = 1500$. The posterior correctly polarises to $\{0, 1\}$ at every off-diagonal entry, with a final Frobenius error of $\|\hat{W} - W^*\|_F \approx 0.41$ on a matrix of Frobenius norm $\sqrt{n(n-1)p_{\text{edge}}}$. Residual fuzziness is concentrated on edges of neurons with low spike rates, for which identifiability is slower (cf. the KL calculation in Section 4.1).

6.4 Scaling of the recovery error

Figure 4 shows the time-evolution of the normalised Frobenius error

$$e(t) := \|\mathbb{E}[W | \mathcal{G}_t] - W^*\|_F / \sqrt{n(n-1)}$$

for $n \in \{3, 5, 7\}$ and 3 i.i.d. replicates each (different W^* and different seeds). Bigger networks require longer observations, both because the number of edges grows like n^2 and because the individual firing rates decrease at fixed prior edge probability. The curves nevertheless exhibit the same qualitative shape: a transient plateau during which the first few spikes are aggregated, followed by a regime where the error decays roughly like $1/\sqrt{t}$, consistent with the CLT prediction of Proposition 4.1.

7 Extension: continuous-valued interactions via SMC

The binary constraint $W_{ji} \in \{0, 1\}$ is convenient for pedagogy but restrictive: in many applications the interaction strengths are graded, and the inverse problem is intrinsically infinite-dimensional [Lu et al., 2019]. When the prior on W admits a Lebesgue density, the posterior p_t is no longer supported on a finite set and the exact filter of Proposition 3.3 becomes intractable. We propose here a simple *bootstrap particle filter* [Doucet et al., 2000, Chopin and Papaspiliopoulos, 2020] that demonstrates, as a proof of concept, that the filtering perspective survives the discrete-to-continuous transition.

7.1 Bootstrap particle filter

We focus on the two-neuron model of Section 3.1 with $J \in \mathbb{R}$ and Gaussian prior $J \sim \mathcal{N}(\mu_0, \sigma_0^2)$. The algorithm maintains M weighted particles $(J^{(m)}, w^{(m)})$; each particle also tracks its own potential $V^{(m)}$ driven by the observed jumps of $\tilde{N}$ and N , because V depends on J . Writing t_{k-1}, t_k for two consecutive events and noting that $V^{(m)}$ is constant between events, the likelihood

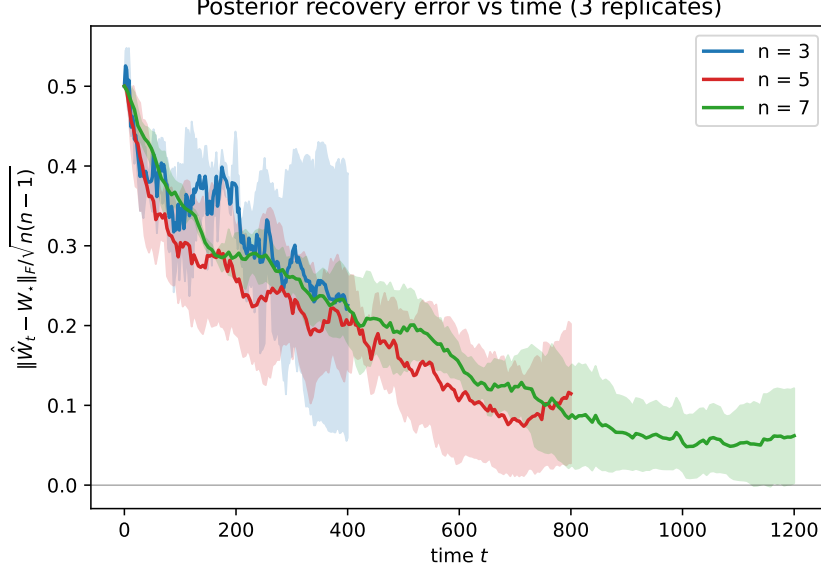


Figure 4: Mean $\pm$ standard deviation of the normalised Frobenius recovery error $\|\hat{W}_t - W^*\|_F / \sqrt{n(n-1)}$ across 3 MC replicates, for $n \in \{3, 5, 7\}$. All replicates show the expected long-time decay of the posterior error.

contributed by that inter-event interval is, for each particle,

$$\Delta \log w_k^{(m)} = -(t_k - t_{k-1}) f(V_{t_{k-1}}^{(m)}) + \mathbf{1}\{\text{event is a spike of } N\} \cdot \log f(V_{t_k}^{(m)}), \quad (18)$$

and in between the $V^{(m)}$'s are updated by $V^{(m)} += J^{(m)}$ at each jump of $\tilde{N}$ and reset to zero at each jump of N . The effective sample size is monitored and whenever it falls below $M/2$ we resample with replacement and apply a small Gaussian jitter $J^{(m)} \leftarrow J^{(m)} + \sigma_k \epsilon_m$, with $\sigma_k = \sigma_0 \cdot (1+k)^{-\alpha/2}$ and $\alpha \in (0, 1)$ small. This is a static-parameter analogue of the perturbation trick of Liu and West [2001] and ensures asymptotic unbiasedness [Chopin and Papaspiliopoulos, 2020, Ch. 17]. Details, including the pseudocode, are given in the accompanying module `smc.py`.

7.2 Proof-of-concept experiment

Figure 5 reports the posterior mean and std of J over time as the filter processes the observed spike train, for three ground-truth values $J^* \in \{0, 1, 2\}$. With $M = 2000$ particles, prior $\mathcal{N}(0.5, 1)$ and $T = 500$, the posterior concentrates sharply on $J^* = 0$ and reasonably on $J^* = 1$. For $J^* = 2$ the filter underestimates J systematically: this is not a bug but a *saturation effect* of the sigmoid. Since $f(v) \rightarrow 1$ as $v \rightarrow \infty$, any $J \gtrsim 2$ produces nearly indistinguishable likelihoods whenever $V_t \geq 2$, hence the likelihood ridge extends to arbitrarily large J and the posterior inherits an almost flat tail (visible in the right histogram). This is a feature shared by any unidentifiable parameter: the Bayesian filter reports genuine uncertainty rather than a misleadingly sharp point estimate, which is arguably the correct behaviour.

Two directions to scale the SMC to the n -neuron setting deserve mention. First, the column-wise factorisation of Lemma 5.1 still holds for continuous W , so one may run n independent SMC filters of dimension $n - 1$ each rather than a single filter of dimension $n(n - 1)$. Second, static-parameter degeneracy can be further mitigated with SMC² or marginalised particle filters [Chopin et al., 2013] or with amortised variational posteriors over W . We leave these to future work.

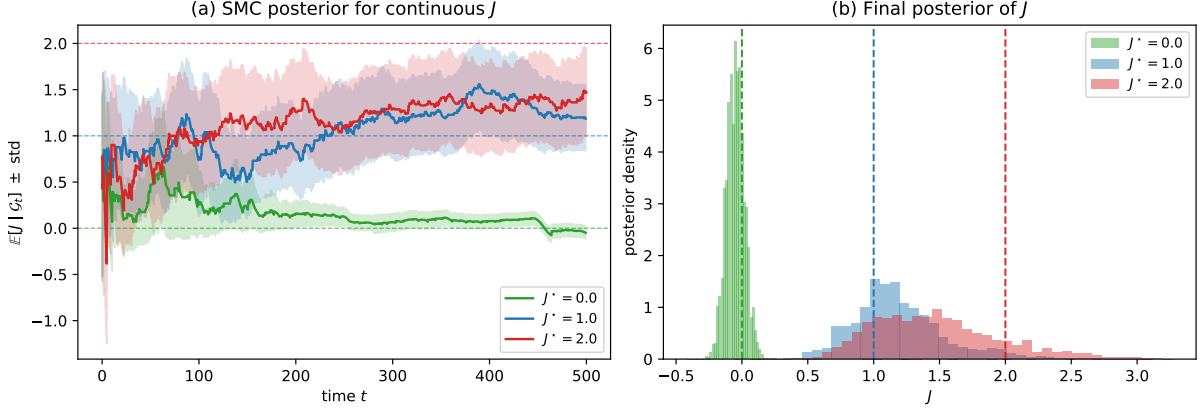


Figure 5: Bootstrap SMC with $M = 2000$ particles on the two-neuron model with continuous J , for ground-truth values $J^* \in \{0, 1, 2\}$. (a) Posterior mean $\pm$ standard deviation over time; dashed lines mark the respective J^* . (b) Final posterior histograms at $T = 500$.

7.3 Interacting particle filter

In this subsection, we justify theoretically the use of a different particle filter based on interacting particles. The algorithm is as follows:

- at $t = 0$, sample M particles in the search space according to a prior distribution p_0 . Each particle $(w^1, \dots, w^M)$ is equipped with a Poisson clock N_t^m with intensity $\sum_i f(V_t^i(w^m)) = \Lambda_t(w^m)$;
- when N_t^m fires, particle w_t^m is replaced by particle $w_t^{m'}$, with m' sampled uniformly among the particle indices $\{1, \dots, M\}$;
- at a jump of neuron i , each particle w^m is independently redrawn from the categorical distribution that puts mass $f(V_{\tau_k^i}^i(w^{m'}) / \sum_{m''} f(V_{\tau_k^i}^i(w^{m''}))$ on $w^{m'}$, for $m' \in \{1, \dots, M\}$.

This algorithm has the advantage of reducing the variance of the posterior by rapidly removing low-likelihood particles. However, it incurs the risk of losing good particles prematurely from the random selection mechanisms. Future work will aim to combine this algorithm with likelihood based SMC methods to get the best of both worlds. Gradient based methods could also be included to provide these algorithms with some exploration abilities.

7.3.1 Generator of the interacting particle filter

We prove in this section that the generator of the Markov chain defined by the interacting particle filter algorithm converges to the general posterior ODE. To achieve this, we first derive the general posterior ODE in the sense of distributions.

For a probability distribution p and test function Φ , we denote

$$\mathbb{E}_p[\Phi(w)] = \int \Phi(w)p(w) dw = p(\Phi)$$

Recall the formula for the posterior ODE between spikes $p_t(w) = \frac{L_t(w)}{\mathbb{E}_{p_0}[L_t(w)]}$

In weak form, this formula becomes

$$p_t(\Phi) = \frac{\mathbb{E}_{p_0}[\Phi(w)L_t(w)]}{\mathbb{E}_{p_0}[L_t(w)]} = \frac{p_0(\Phi L_t)}{p_0(L_t)} = \frac{\gamma_t(\Phi)}{\gamma_t(1)}$$

Between spikes, the time derivatives of the unnormalised quantities are $\dot{\gamma}_t(\Phi) = -\gamma_t(\Lambda_t \Phi)$ and $\dot{\gamma}_t(1) = -\gamma_t(\Lambda_t)$. The quotient rule then yields

$$\begin{aligned} \dot{p}_t(\Phi) &= \frac{\dot{\gamma}_t(\Phi) \gamma_t(1) - \gamma_t(\Phi) \dot{\gamma}_t(1)}{\gamma_t(1)^2} \\ &= -\frac{\gamma_t(\Lambda_t \Phi) \gamma_t(1) - \gamma_t(\Phi) \gamma_t(\Lambda_t)}{\gamma_t(1)^2} \\ &= -\frac{\gamma_t(\Lambda_t \Phi)}{\gamma_t(1)} + \frac{\gamma_t(\Phi) \gamma_t(\Lambda_t)}{\gamma_t(1)^2} \\ &= p_t(\Phi) p_t(\Lambda_t) - p_t(\Lambda_t \Phi). \end{aligned}$$

At a jump of neuron i , the update in weak form writes

$$p_{\tau_k^i}(\Phi) = \frac{p_{\tau_k^i-}(f(V_{\tau_k^i-}^i(w))\Phi)}{p_{\tau_k^i-}(f(V_{\tau_k^i-}^i(w)))}$$

We now turn to the particle filter equations. Let $W = (w^1, \dots, w^M)$ denote the population vector. The generator of the particle filter $\mathcal{L}$ splits into a continuous-time component $\mathcal{L}_{\text{cont}}$ and a jump component $\mathcal{L}_{\text{spike}}$:

$$\begin{aligned} \mathcal{L}_{\text{cont}}\Phi(W) &= \sum_m \Lambda(w^m) \frac{1}{M} \sum_{m'} [\Phi(W^{m \rightarrow m'}) - \Phi(W)], \\ \mathcal{L}_{\text{spike}}\Phi(W) &= \sum_i f(V^i(w^*)) \int [\Phi(\tilde{W}) - \Phi(W)] \prod_m q_W^i(d\tilde{w}^m), \end{aligned}$$

where w^* is the true (latent) connectivity matrix driving the observed spikes (so neuron i fires at rate $f(V^i(w^*))$), $W^{m \rightarrow m'}$ is the population vector with w^m replaced by $w^{m'}$, and

$$q_W^i(d\tilde{w}) := \sum_m \delta_{w^m}(d\tilde{w}) \frac{f(V^i(w^m))}{\sum_{m'} f(V^i(w^{m'}))}$$

is the likelihood-reweighted empirical measure for neuron i . In words: at each spike of the true process driven by w^* , the M particles are jointly resampled, each $\tilde{w}^m$ being drawn independently from q_W^i , in line with the algorithm described above.

We apply these generators to the test function $\Phi(W) = \frac{1}{M} \sum_m \phi(w^m) = p^W(\phi)$ defined by the empirical measure $p^W = \frac{1}{M} \sum_m \delta_{w^m}$. For the continuous part,

$$\begin{aligned} \mathcal{L}_{\text{cont}}p^W(\phi) &= \sum_m \Lambda(w^m) \frac{1}{M^2} \sum_{m'} [\phi(w^{m'}) - \phi(w^m)] \\ &= \frac{1}{M^2} \left[\sum_m \Lambda(w^m) \sum_{m'} \phi(w^{m'}) - M \sum_m \Lambda(w^m) \phi(w^m) \right] \\ &= \left(\frac{1}{M} \sum_m \Lambda(w^m) \right) \left(\frac{1}{M} \sum_m \phi(w^m) \right) - \frac{1}{M} \sum_m \Lambda(w^m) \phi(w^m) \\ &= p^W(\Lambda) p^W(\phi) - p^W(\Lambda \phi), \end{aligned}$$

which recovers the continuous-time posterior ODE.

For the spike part, since $p^{\tilde{W}}(\phi) = \frac{1}{M} \sum_m \phi(\tilde{w}^m)$ and the $\tilde{w}^m$ are i.i.d. draws from q_W^i ,

$$\begin{aligned} \mathcal{L}_{\text{spike}} p^W(\phi) &= \sum_i f(V^i(w^*)) \int [p^{\tilde{W}}(\phi) - p^W(\phi)] \prod_m q_W^i(d\tilde{w}^m) \\ &= \sum_i f(V^i(w^*)) \left[\int \phi(\tilde{w}) q_W^i(d\tilde{w}) - p^W(\phi) \right] \\ &= \sum_i f(V^i(w^*)) \left[\frac{p^W(f(V^i)\phi)}{p^W(f(V^i))} - p^W(\phi) \right], \end{aligned}$$

which recovers the Bayesian spike update of Proposition 3.3. Additional mean-field arguments are required to prove the convergence of p^W to p as $M \rightarrow \infty$, which is left for upcoming work.

8 Discussion and outlook

Relation to targeted learning. The interaction-structure inference problem was discussed in Lecture 6 of the course through the targeted-learning framework of van der Laan and Rose [2011], applied to Ising-type couplings via odds-ratio estimation. That approach is agnostic about the dynamics and operates on equilibrium snapshots. Our filter is complementary: it requires a fully specified point-process generative model, but in exchange yields posterior consistency together with explicit non-asymptotic rates (Eq. (11) and Proposition 4.1), as well as the full posterior rather than a point estimate. When the dynamics are well-identified from the first principles of the system (as in spiking networks or epidemic contacts), our filter provides sharper uncertainty quantification; when only weak structural assumptions are available, targeted learning remains the natural tool.

Relation to ABM inference and Expectation–Maximisation. Agent-based-model inference (Lecture 6; Banisch, 2015) typically treats the interaction parameters as latent and optimises them by EM on time-aggregated observables. In our setting, the “E-step” is replaced by the closed-form filtering formulas of Proposition 3.3, which integrate the continuous-time likelihood exactly rather than its time-averaged moments; no iteration is needed, and the posterior is available at every observation time. A related advantage is the ability to start a new prior at any time without re-optimising: the filter can be warm-started from its current posterior, which suggests natural extensions to streaming settings or online hyperparameter adaptation.

Causal interpretation. Because W encodes a directed interaction graph, the posterior p_t^W defines a distribution over causal structures in the sense of Pearl [2009]. The filtering viewpoint therefore provides a continuous-time, model-based route to causal discovery. We highlight two important caveats discussed in the course’s reading of Manzo [2022]. First, identifiability of W requires the intensity f to be non-constant over the reachable set of potentials; a flat f makes W causally underdetermined. Second, the filter is honest about unidentifiability (as shown by the flat posterior tail in Figure 5 for $J^* = 2$), which is a meaningful positive when compared to point-estimate-based methods that may give misleadingly precise answers.

Connections with rare events (Bloc 4). The consistency argument of Section 4.1 is a large-deviation-type statement on the log-likelihood ratio. We conjecture that the posterior of W obeys a Freidlin–Wentzell-style LDP with rate function governed by the KL divergences between inter-event laws across hypotheses, bridging the inference problem with the rare-event themes of the course. A quantitative statement of this type would sharpen Proposition 4.1 and connect it to the extreme-event literature on point processes.

Limitations and future work. The filter currently assumes the intensity f to depend only on the immediate past potential of the neuron, and the column-wise consistency sketch of Section 5 additionally assumes $f \geq f_{\min} > 0$, an assumption violated by the sigmoid intensity used in Section 6 (see the remark at the end of Section 5). Extensions to intensity functions with memory kernels (e.g. Hawkes processes) would broaden applicability. Moreover, the current SDE driving the network is relatively simple, and more detailed biological features will almost necessarily be required for robust applications of the model to real data. However, the current SDE can be easily modified in certain ways without altering the core methodology, such as by introducing a potential drift between jumps. The combinatorial cost $n \cdot 2^{n-1}$ still scales exponentially in n and therefore caps exact inference at $n \sim 10$; graph-sparsity priors and variational approximations are natural ingredients for further scaling. The SMC extension can be embedded in the column-wise factorisation to handle medium-sized continuous- W networks; a systematic study of degeneracy and resampling schedules is left for future work.

Reproducibility. All figures in the paper can be reproduced from the accompanying Python code by running the scripts `code/experiments/fig{1,2,3,4,5}_*.py`. Runtime on a laptop is under two minutes in total after caching.

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A Sketch of the filter derivation

Let $\mathbb{P}^0$ be a reference measure under which N is a Poisson process of rate $f_{\max}$ independent of everything else; J and $\tilde{N}$ retain their prior laws. Under $\mathbb{P}^0$ the filter is trivial because N is uninformative. The Radon–Nikodym derivative between the true law $\mathbb{P}^J$ and $\mathbb{P}^0$ on $\mathcal{G}_t$ is the Doléans–Dade exponential

$$L_t^J = \exp \left(\int_0^t \log \frac{f(V_s^{(J)})}{f_{\max}} dN_s - \int_0^t [f(V_s^{(J)}) - f_{\max}] ds \right),$$

which corresponds to thinning the Poisson process with rate $f_{\max}$ at rate $f(V^{(J)})/f_{\max}$. By the Kallianpur–Striebel formula, $p_t^1 = \mathbb{E}^0[L_t^1 \mathbf{1}\{J = 1\}]/\mathbb{E}^0[L_t^1 \mathbf{1}\{J = 1\} + L_t^0 \mathbf{1}\{J = 0\}]$, and differentiating in time gives (6) between jumps of N and (7) at jumps of N . Jumps of $\tilde{N}$ contribute equally to L_t^1 and L_t^0 (the rate θ is independent of J) and therefore do not update the posterior. The complete derivation, including the n -neuron case, is given in Section 5.1.

B CLT convergence rate

The random walk (11) has i.i.d. centred increments $X_\ell - \mu$ with finite variance σ^2 . A Berry–Esseen inequality gives, for every $k \geq 1$, $\sup_{x \in \mathbb{R}} |\mathbb{P}((S_k - k\mu)/(\sigma\sqrt{k}) \leq x) - \Phi(x)| \leq C\rho/(\sigma^3\sqrt{k})$, with $\rho = \mathbb{E}|X_1 - \mu|^3$ and C an absolute constant. Proposition 4.1 then follows by noting that $\{k_\varepsilon \leq k\} \supseteq \{S_k \geq A_\varepsilon\}$ under $J^\star = 1$ (the argument is symmetric for $J^\star = 0$).